

Jacopo Zacchigna

Master's student in Data Science & AI

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Summary

Master's student in Data Science & Artificial Intelligence with research experience in mechanistic interpretability, computer vision, and generative models. Strong foundation in statistics, mathematics, and computer science. Published computer vision work in *Plants*. Interested in interpretability, advancing deep learning theory and applied research in real-world systems and HPC.

Education

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| <p>M.S. University of Trieste, Data Science and Artificial Intelligence</p> <ul style="list-style-type: none"> • Probabilistic ML, Advanced Deep Learning, NLP, RL, Advanced HPC, XAI, Theory of Deep Neural Networks, Unsupervised | <p>Trieste, Italy</p> <p>Sept 2024 – Sept 2026</p> |
| <p>B.S. University of Trieste, Artificial Intelligence and Data Analytics</p> <ul style="list-style-type: none"> • Built strong foundations in math, statistics, machine learning, and core CS, with experience in model training and evaluation | <p>Trieste, Italy</p> <p>Sept 2021 – July 2024</p> |

Projects & Outside Experience

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| <p>Monitoring and Attributing Implicit Personalization in Conversational Agents, Research Fellow</p> <ul style="list-style-type: none"> • Conducted experiments on persona-vectors extracted from activations of rich persona (SynthPersona) using NNsight and nnterp • Worked on evaluation for Doc-to-Lora for Synthetic Persona on templated attributes vs full biography • Created an interactive UI with support for persona-vector extraction, analyses | <p>Remote</p> <p>Feb 2026 – May 2026</p> |
| <p>BayesianFlow — Uncertainty Estimation in Generative Models, Personal Research Project</p> <ul style="list-style-type: none"> • Extended BayesDiff from diffusion to flow matching models, achieving 5× faster generation than DDIM with comparable quality • Integrated Last Layer Laplace Approximation into U-Net for pixel-wise uncertainty estimation, producing interpretable confidence maps • Conducted comparative analysis of flow matching and diffusion models, assessing computational efficiency and uncertainty propagation | <p>June 2025 – July 2025</p> |

Professional Experience

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| <p>Obloo, AI Consultant</p> <ul style="list-style-type: none"> • Mentored 5 interns in AI research & engineering practices, leading a 50% increase in team productivity • Designed GenAI Retrieval-Augmented Generation (RAG) systems for knowledge retrieval from company data lake • Led ML research prototypes bridging academic methods with production workflows, contributing to 2 new products adopted company-wide | <p>Trieste, Italy</p> <p>May 2024 – Jan 2025</p> |
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Skills

Skills: Deep Learning, Generative Models, Bayesian Inference, Research, Python, C/C++, Computer Vision, Data Science, Linux/Unix, Natural Language Processing (NLP), PyTorch, R, Docker, High Performance Computing (HPC), ML, Machine Learning, data mining, algorithm development, NumPy, CUDA, GPU

Languages: Italian, English